



Assignment 01

Language Modeling for Urdu News Articles

Natural Language Processing

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Project Overview

This project implements language learning models for BBC Urdu news articles. The models evaluated in this project are unigram, bigram and trigram models which are trained to learn word sequence probabilities and to generate coherent Urdu articles. All modeling is implemented from scratch using basic Python data structures. The goal of this project is to analyze bigram and trigram perplexity to evaluate which model's performance excelled.

Methodology

Data Collection

300 complete news articles related to the genre of sports were scrapped from the BBC Urdu news website. The article metadata, which included the title and publish date, were stored in a separate JSON file. The raw text is saved as in a normal plain file with indexing for each article.

Preprocessing Methods

To preprocess the scrapped data, diacritics, URLs, numbers, emojis and all other non-Urdu or latin alphabet characters were removed. After this, the data is segmented into sentences and whitespace and formatting is normalized. The normalized data is saved as a plain text file separately. Before language model training, the data is further preprocessed. Custom stemmer, lemmatizer and tokenizer are applied for Urdu to divide the text into stems, lemmas and tokens respectively. A dataframe is made to save all tokens, lemmas and stems. The vocabulary of the whole corpus is 11947.

Language Model Training with Smoothing

The language models used are unigrams, bigrams and trigrams. These models were trained on the processed corpus lemmas. To address the challenges of unseen n-grams and the rich and large vocabulary inherent to Urdu, K-smoothing was applied to unigrams and bigrams, while linear interpolation was employed for trigrams to combine the unigram, bigram and trigram probabilities. This hybrid smoothing strategy balances context utilization with probability stability.



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Text Generation Process

Multiple seed phrases were used to generate 3 articles each using bigrams and trigrams to compare both models. Trigrams was also used to generate 5 unique news headlines related to sports. Constraints were enforced to prevent excessively long or incomplete sentences, to ensure the generated text was syntactically coherent.

Evaluation

Model performance was assessed by perplexity on 5 unseen test articles, which were also scrapped from BBC Urdu News website and pertained to sports news to match the lexical theme. Both mean and median perplexity metrics were computed to provide a more robust evaluation and to mitigate outlier influence.

Results

The perplexity results showed that the trigram model performed better than bigram model consistently, which highlights the importance of extended context. Perplexity variation across articles reflects differences in lexicon and article lengths, which is to be expected for n-grams applied to morphologically rich languages like Urdu in this case.

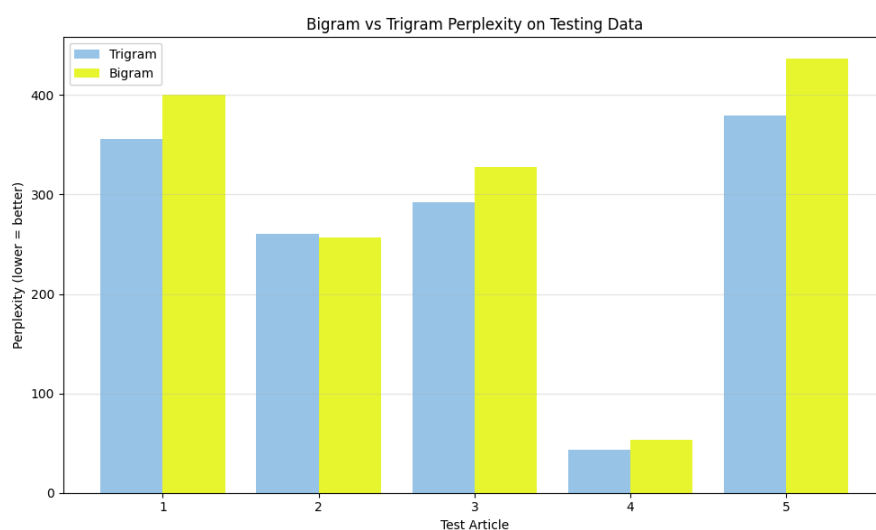
Both trigram and bigram language models were evaluated on five samples. For the trigram model, the perplexity scores obtained were 355.41, 260.62, 292.49, 43.72, and 379.55. The bigram model, on the other hand, produced perplexity values of 399.82, 256.97, 327.47, 53.70, and 436.28 for the same samples.

In terms of overall performance, the trigram model showed a lower average perplexity of 266.36 and a median of 292.49. The bigram model had a higher mean perplexity of 294.85 with a median value of 327.47, suggesting that the trigram model performed better overall.

The detailed values are given in the Table and Figure below.

Sample	Trigram Perplexity	Bigram Perplexity
1	355.41	399.82
2	260.62	256.97

3	292.49	327.47
4	43.72	53.70
5	379.55	436.28
Mean	266.36	294.85
Median	292.49	327.47



Discussion

Challenges and Key Findings

The findings of this project demonstrate that higher order n-gram models like the Trigram model substantially enhance predictive accuracy, especially when they are paired with smoothing techniques. The project also demonstrates that n-gram models can generate articles and other text based on a small corpus. However, there are several limitations and



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challenges in this project. Firstly, the dataset was very limited so while the vocabulary size was high, the frequency of words was small. This led to common words having minimal probabilities in unigrams and sparse data in general for all models. While this project highlights the effectiveness of classical statistical models in low-resource languages and constraints, it provides a foundation for further research and exploration of probabilistic models in Urdu.

Conclusion

In conclusion, this project implemented n-gram modeling on low-resource language, Urdu, to generate Urdu news articles and compare performance of models, without external libraries. This work highlights the viability of using web scraping tools to extract data for low-resource languages, and building a preprocessing pipeline from scratch in order to align to the needs of the language. Due to the morphological richness of Urdu and the small dataset used, perplexity values were on average between 300-400. Further work is suggested to apply more advanced statistical models on larger datasets.